

Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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| The puzzle

- ◆ Polarization should make legislatures **ungovernable**.
- ◆ Brazil 2014–16: impeachment · Lava Jato · recession · polarization.
- ◆ Yet the Chamber **kept forming majorities**, across impeachment and a change of power.
- ◆ The literature's crisis typology: **collapse · fragmentation · realignment · institutionalization**. Brazil fits **none**.
- ◆ **How? And at what cost?**

Coalitional presidentialism, and a cluster of shocks

- ◆ Strong president, **no majority** → govern by **coalition** (Abranches 1988).
- ◆ Levers: **budget · posts · agenda**; the **Centrão** sells support as the president's *agent*.
- ◆ 2014–16, all at once: impeachment · Lava Jato · recession · and the president's budget lever removed: **individual amendments made mandatory** (*emendas impositivas*, 2015).
- ◆ **The shocks are collinear** → no single lever can be isolated (hence the eliminative design).

| The original design was causal

- ◆ The plan was causal: estimate whether the 2015 mandatory-amendment reform (*emendas impositivas*) reshaped coordination, with a **temporal ERGM**.
- ◆ The obstacle is **identification**: the shocks are collinear, so a single effect cannot be separated.
- ◆ A first bipartite ERGM also would not mix, but that is secondary; even a clean fit would face the same identification problem.
- ◆ So the design **eliminates the rival explanations** one at a time, rather than estimating one effect.

| The structural question

- ◆ The open question is structural: **how did the chamber's coordination reorganize after the 2014–2016 shocks, and what happened to its pragmatic center?**
- ◆ And, as the puzzle set up: **at what cost?**

The tests that follow: behavior not composition, durable, robust to the rival explanations.

| The thesis, and what it means

- ◆ The pragmatic center has become the recurring pivot through which governments build majorities.
- ◆ Coordination by **position relative to the executive**, beyond a fixed ideological axis.
- ◆ The Centrão is **old** (1987); its **position** changed: *agent* → *pivot*.
- ◆ What it means: governability **sold, not represented**; the cost is **accountability**.

| Why networks, not ideal points

- ◆ **deputies × bills** → agreement graph → **communities**.
- ◆ Standard **one-dimensional ideal-point** models compress change onto one axis.
- ◆ Blocs **recompose** while ideology looks stable; an axis can't see it.
- ◆ No axial assumption → the shift is **bloc recomposition**.

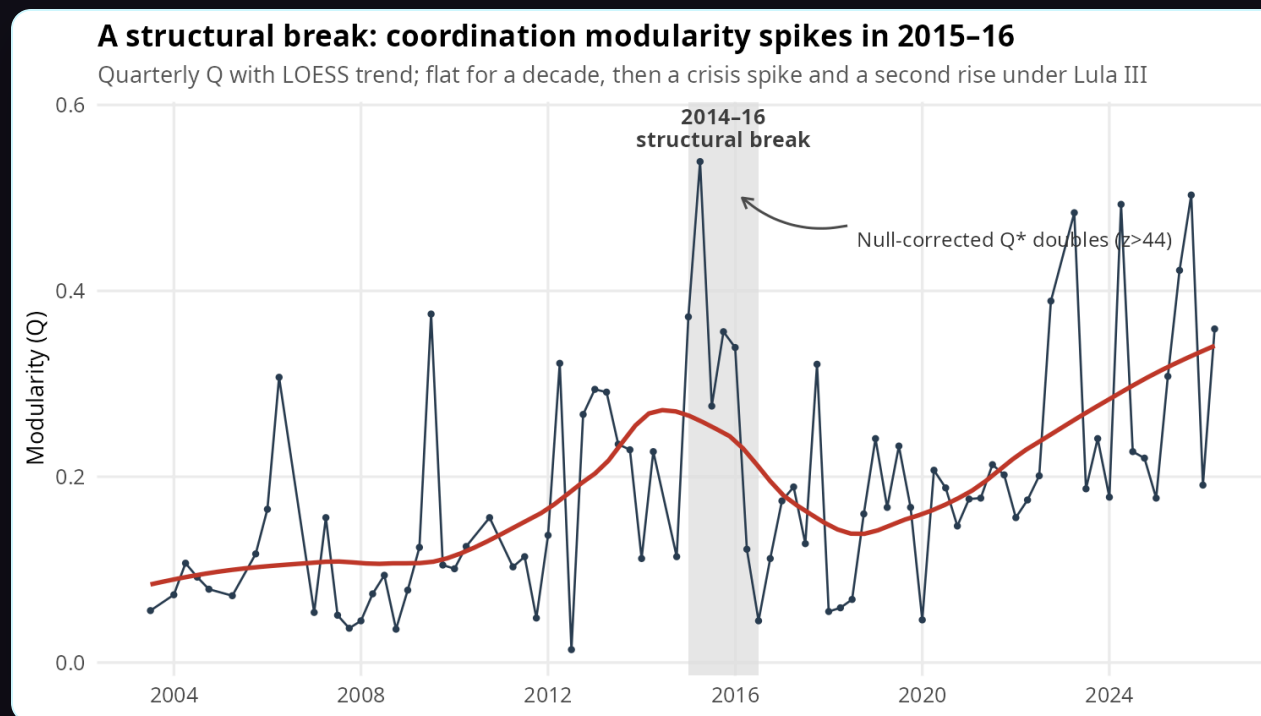
| The methodological approach

- ◆ **Measure:** roll-calls → **deputy × bill** graph → **communities per legislature**.
- ◆ **Classify bills:** fiscal vs moral, via a local **LLM (Qwen 2.5 32B)**, human-validated (n=294, Krippendorff's $\alpha=0.84$, 89.5%).
- ◆ **Compare:** across the 2014–16 break; a **within-deputy panel** separates behavior from turnover.
- ◆ **Falsify, not estimate:** collinear shocks → **eliminate rivals one at a time**.
- ◆ **Stress-test:** Q^* null, 12-cell threshold grid, alternative projections and algorithms.

| Related work, and how this differs

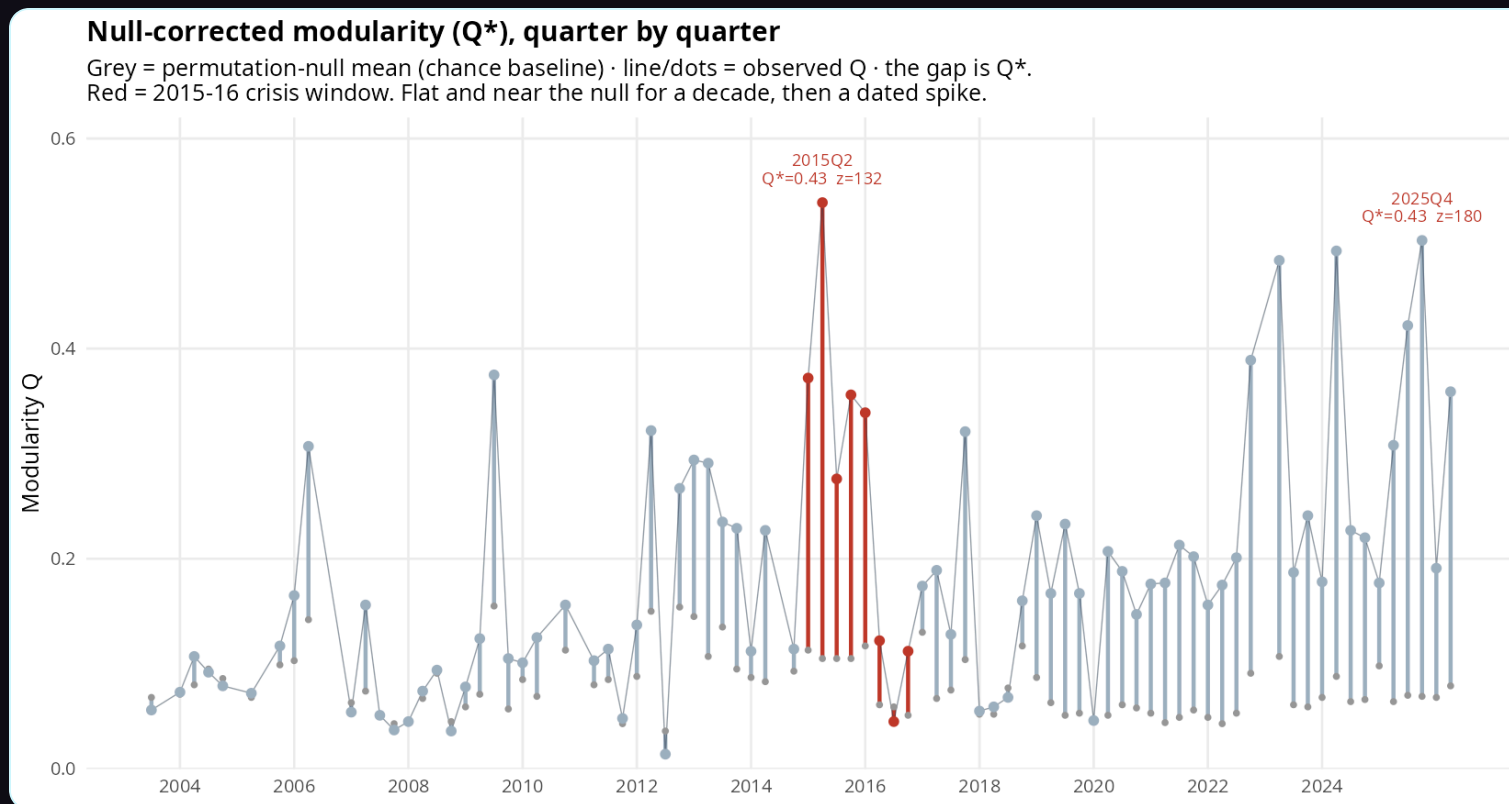
- ◆ **Networks on the Chamber already exist** (Brito 2020; Toledo Jr. 2025): voting communities, an isolation spike at the 2016 impeachment.
- ◆ **This paper differs in four ways.** It **begins from the bipartite** deputy×bill incidence and tests **alternative projections** and a **projection-free SBM**.
- ◆ Modularity is benchmarked against a **within-vote permutation null (Q^*)**; a **within-legislator** panel separates behavior from composition.
- ◆ And it is read through **coalitional presidentialism**: the Centrão **pivot**, not polarization, is what is measured.

A structural break: modularity spikes in 2015–16



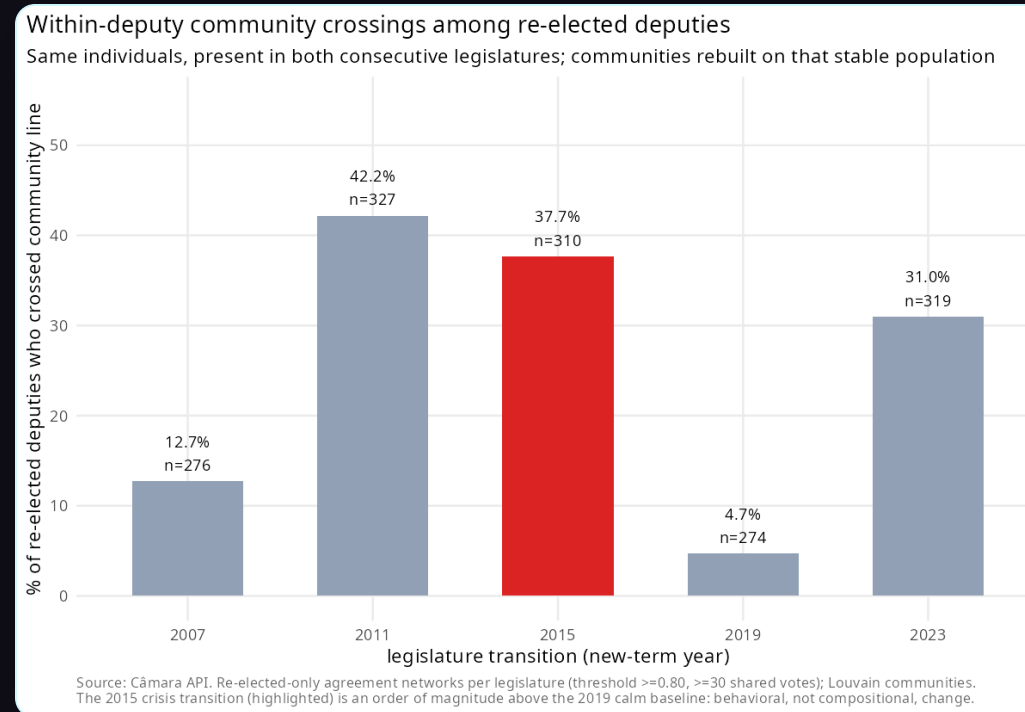
Quarterly Q: flat ~0.15, spikes to 0.51 in 2016. Null-corrected Q^* doubles at the 55L and 57L.

Null-corrected modularity (Q^*) over time



Quarterly observed Q rises far above a flat permutation-null baseline. Q^* peaks at the 2015–16 crisis ($z \approx 132$) and rises again under Lula III; the null stays low throughout.

Same people, not turnover

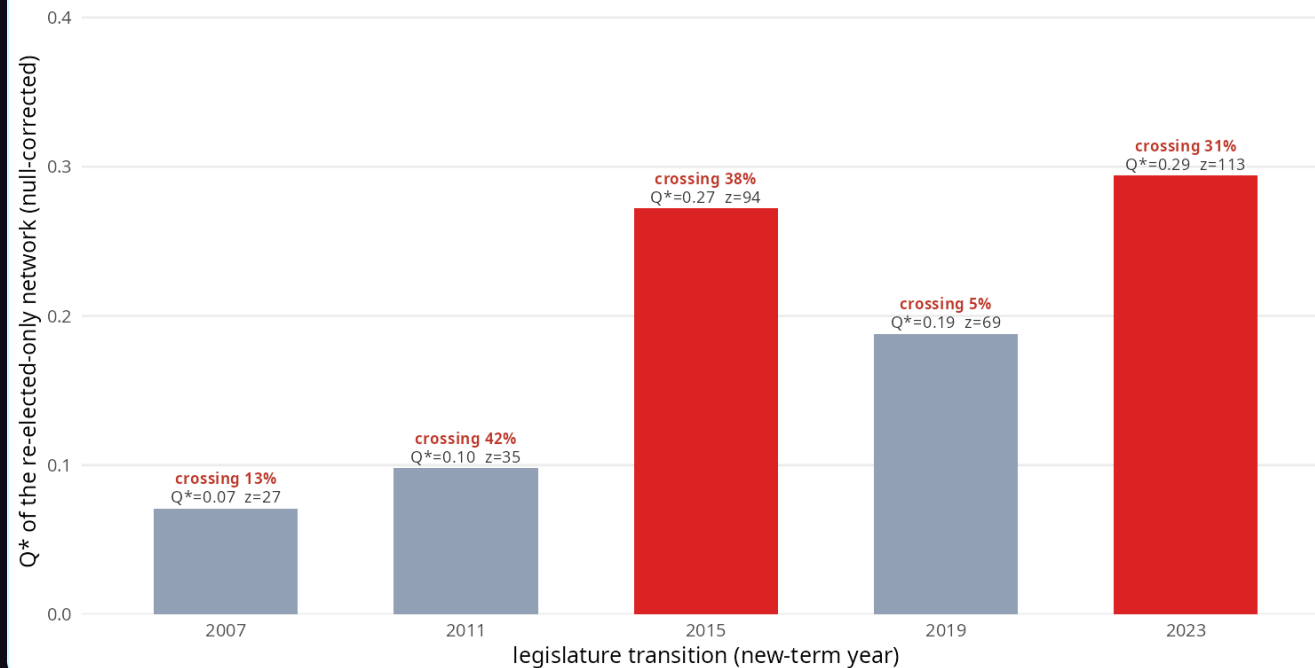


Reelected deputies crossing communities: **37.7%** at the break vs **4.7%** after. **The re-sorting is among the same legislators, so it is not legislative turnover.**

Movement is not crystallization

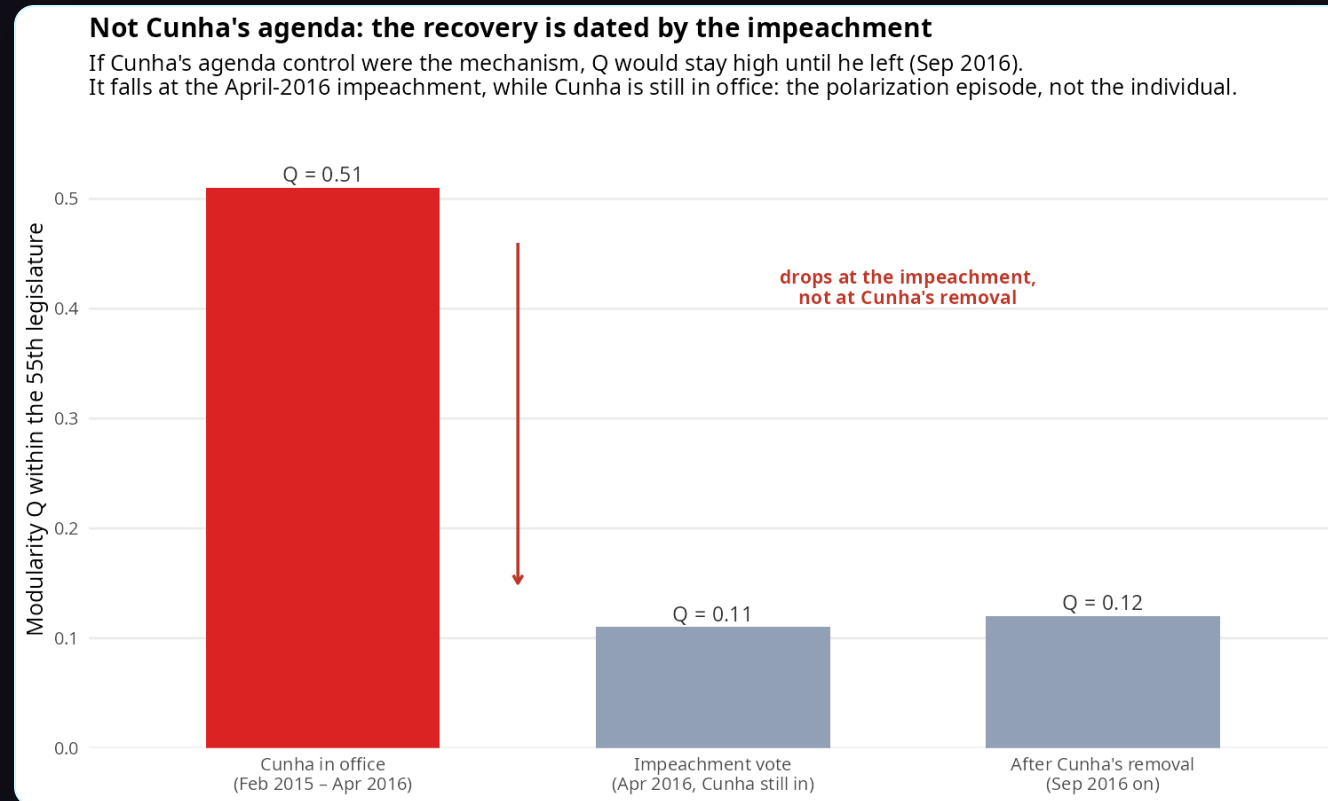
Same people: movement vs crystallization

Null-corrected modularity among deputies present in both legislatures. Crossing is movement, Q^* is structure. The two re-sorts (2015 crisis, 2023 Lula III alternation) move the same people AND sharpen the structure; 2011 churns (high crossing, low Q^*) and 2019 holds (consolidated, little movement).



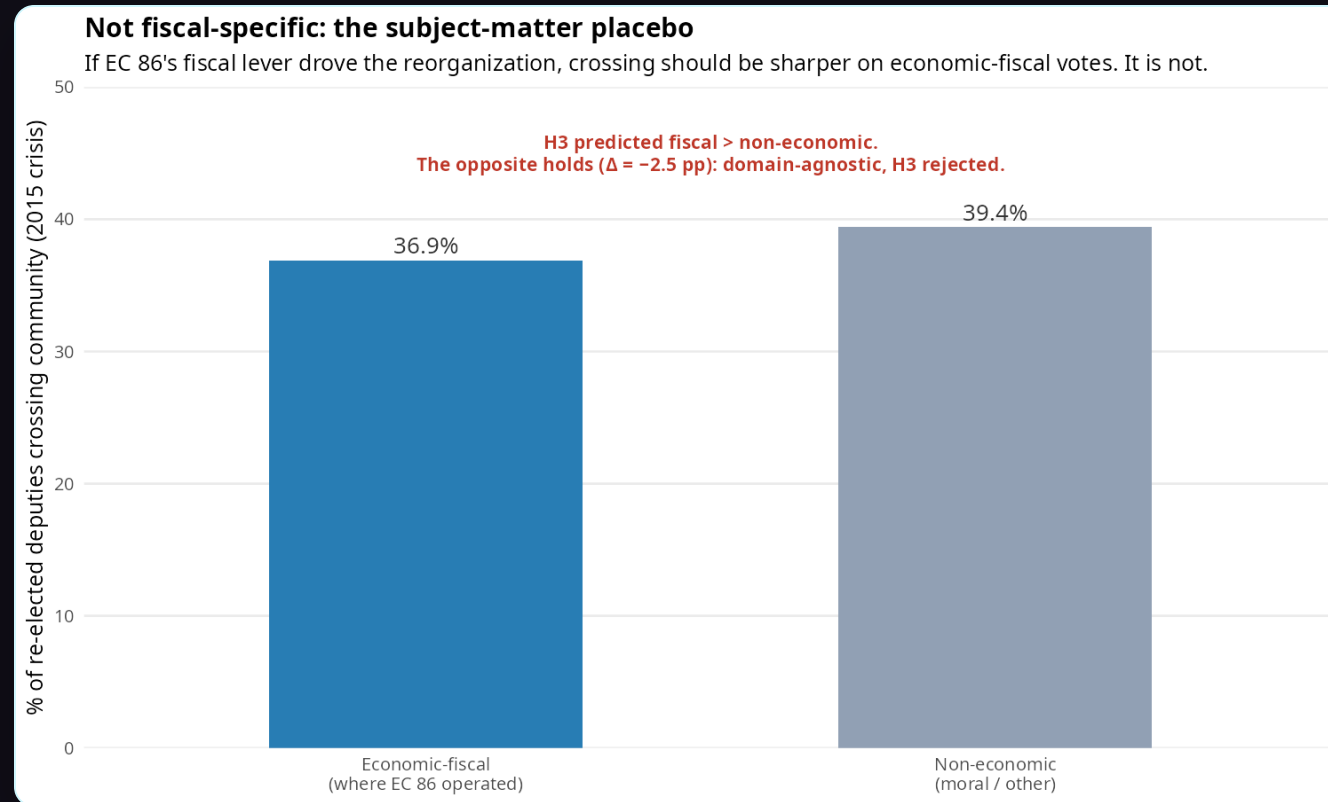
Same re-elected deputies: 2011 moves more (42% crossing) but stays diffuse (0.10); the 2015 crisis sharpens the structure (0.27) and it persists.

Not Cunha's agenda: dated by the impeachment



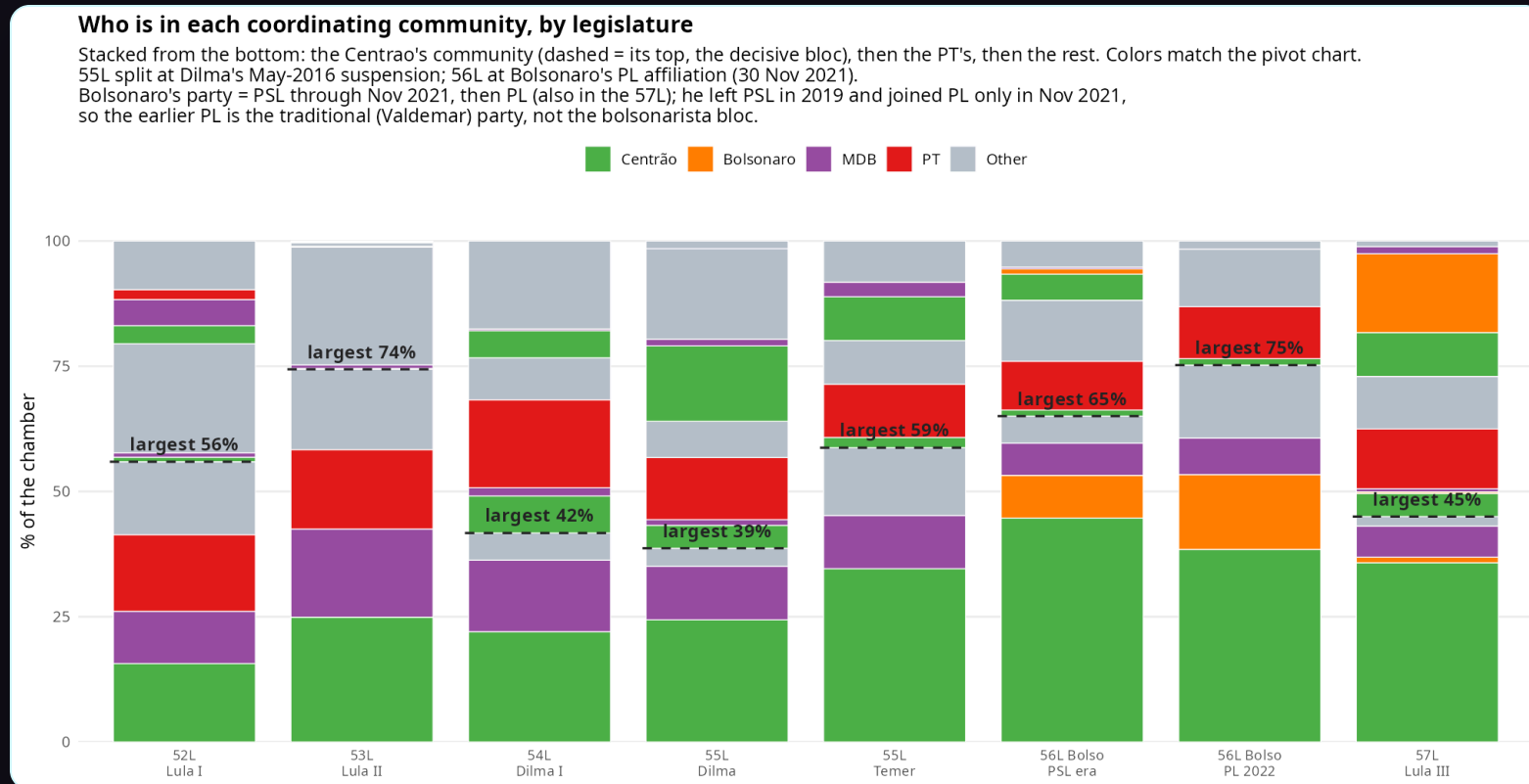
Within the 55L, modularity falls at the April-2016 impeachment, not at Cunha's September-2016 removal. Dropping urgency votes moves the crossing 0.1 pp; a Lira-presidency analog shows no spike.

Not fiscal-specific: the subject-matter placebo



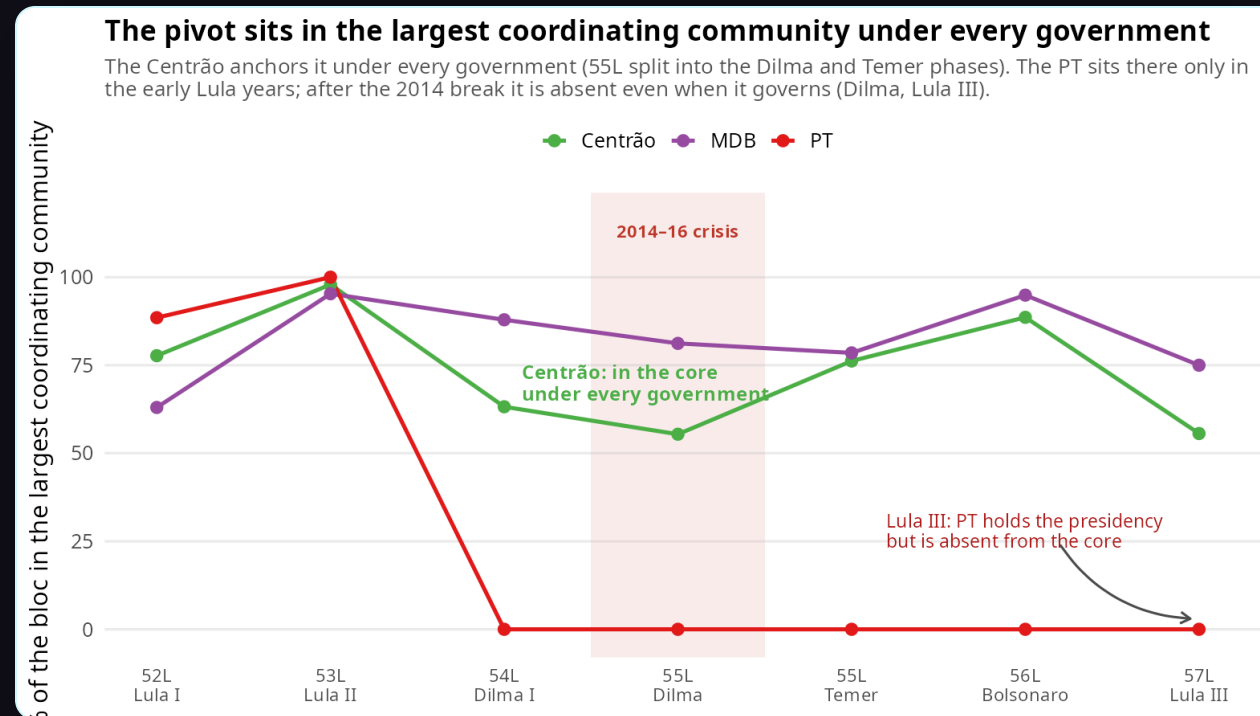
Crossing is no larger on economic-fiscal votes than on non-economic ones ($\Delta = -2.5$ pp), robust to procedural coding and to a null pre-2015 dose-response. H3 rejected: the disruption is systemic, not sectoral.

The coordinating communities, sized



Few communities (3–4); the largest is big enough to decide outcomes. From the 2014 break the Centrão anchors the largest bloc and the PT falls to a smaller one. Bolsonaro's party (orange) is inside the largest while he governs (56th) and moves out under Lula III; it is counted as PSL through his PL affiliation (Nov 2021) and PL after, so the earlier PL is the traditional party.

The pivot: in the largest community under every government



The Centrão is in the largest coordinating community under every government. The president's programmatic party (PT) sits there only under Lula I-II; from the 2014 break it is absent even when it governs (Dilma, Lula III).

| From agent to principal, and the cost

- ◆ **Why the position flips (a reading, not a tested cause):** the 2014–16 rupture began the transfer to Congress (**mandatory amendments**); later instruments, the **secret budget** and **strategic agenda control**, consolidated it, so the tool that once disciplined the pivot now lets it **set the terms** (*governo congressuale*).
- ◆ Governability comes to be **sold, not represented**.
- ◆ Cost = **accountability**, not gridlock: a bloc available to any government is hard to hold to account.

| Contribution, and what the case teaches

- ◆ **Measurement:** recover the Centrão's permanence from the votes as a longitudinal quantity, its position and persistence, where the literature had narrative.
- ◆ **Design:** with collinear shocks, eliminate the rival explanations rather than estimate one effect.
- ◆ **A way of looking, not a new claim:** the permanent-pivot type is old (Israel, post-war Italy).

Thank you

Questions welcome.

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